

Boundary tones as sandhi triggers in Xiamen-Taiwanese

Andrew Nevins (Univ. College London) & Nicholas Rolle (Princeton)*

Abstract

In this chapter, we revisit tone sandhi in the Xiamen-Taiwanese cluster of Southern Min. Xiamen-Taiwanese has a seven-way tone contrast where each tone has two forms, one found in phrase-final position (which includes words in isolation) and one in non-final position. Traditionally, the form in final position is taken as the basic form (i.e. underlying), from which the non-final form is derived through its famous tone circle ($24 \rightarrow 33 \rightarrow 21 \rightarrow 52 \rightarrow 44 \rightarrow 33$). This set of alternations is frequently characterized as unnatural and underivable in formal phonology, and is instead said to require a special para-phonological process of ‘paradigmatic replacement’. We argue against this traditional analysis, and instead interpret the non-final forms as the basic forms from which final forms are derived. In our analysis, we decompose the seven tones into tonal features [H] and [L] and register features [*h*] and [*l*], rendering the phonological representation distinct from a tone’s surface pitch. To derive the final forms of words, we posit that each phonological phrase ends in a polar register boundary tone (abbreviated $\hat{r}$), which surfaces with the opposite value of the register of its base. By positing this one single phonological unit ($\hat{r}$), we can account for all tonal alternations. In total, under our analysis the sandhi system is far from arbitrary, underivable, or even peculiar to Xiamen-Taiwanese: it is a well-behaved tonal system involving pre-established notions of tonal features, boundary tones, polarity, and markedness repairs (e.g. spreading and delinking).

1. Introduction

This paper examines Chinese tones, specifically the phenomenon of tone sandhi in Xiamen-Taiwanese (abbreviated as XT) whereby tones change their value depending on whether they are in the final or non-final position of a phrasal constituent. Naively, there are two basic approaches: the *final-as-basic* view takes the final form as more or less underlying and derives the non-final form from it, while the *non-final-as-basic* view take the opposite

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position. In the vast majority of XT sandhi literature (especially in the last 30 years), the final-as-basic view has prevailed. We contend that this is a mistake, though an understandable one. A word pronounced in isolation is incidentally phrase-final, thereby privileging the final-as-basic view. It may be natural for literate societies to take the ‘isolation’ form of a word as the most basic one, even though this may not be the most naturalistic use of it in real-world dialogue. This kind of analytic ‘fallacy’ (not *always* fallacious of course, but more often than is healthy) is a general issue for underlying representations (see e.g. discussion in Albright 2002, and prior extensive foundational discussion in Kenstowicz and Kisseberth 1977).

Instead, we contend that the non-final-as-basic position is essentially correct, a position which allows a much more natural integration with analytic issues of tonal markedness, directionality of tone alternations, and the morphology-phonology interface. This position requires a more abstract phonological representation than what is traditionally adopted for XT tones (i.e. going beyond Chao numbers). Here, we follow a 21st century turn in phonological theory which takes phonological representations as having the potential to be more articulated and abstract than their phonetic realizations (e.g. Trommer and Zimmermann 2014, Trommer 2015, 2024, Zimmermann 2017, *inter alia*).

The XT pattern in particular has received more attention than many other tone sandhi patterns because of its ‘circle’ shape or, more accurately, ‘lollipop’ shape, in which tone A becomes B, B becomes C, C becomes D, and D becomes A again, all of which make it impossible to state transitively closed markedness relations among these tones, given a circular pattern). Additionally, there is a ‘tail’ in which tone E, never an output of the circle, also becomes A. Given the impossibility of dealing with these mappings purely within a markedness scale such as $*A > *B > *C > *D$, certain theorists – particularly in Optimality Theory (Moreton 1999), though even decades earlier in Hsieh (1970) – have denied the psychological reality of the circle, or proposed allomorphy-based treatments with no phonological grounding. As it turns out, to a large part the geometry of the lollipop shape is an artifact of the nature of tone ‘A’ in the above schema. Specifically, when viewing ‘A’ (or any other part) as the phonetic neutralization of distinct phonological representations, the circle is broken. In other words, tones which are identical phonetically may be distinct phonologically (a position with much precedence in the Sinitic tone literature – e.g. Bao 1999).

Against this backdrop, we buck tradition and adopt the non-final-as-basic

view, whereby XT tones change from their underlying form *only* when they are in final position. What conditions this change? Here, we exploit a widely accepted prosodic unit which occurs at the edges of prosodic phrases: the *boundary tone* (Lieberman 1975, Goldsmith 1978, Pierrehumbert 1980). Such boundary tones are not associated with any specific lexical item, but are automatically generated at the end or beginning of a post-lexical constituent. When boundary tones and lexical tones co-occur, certain markedness violations occur which must be repaired via various tonal processes (e.g. spreading and delinking, among others). This is the crux of our argument: the tonal alternations are the result of lexical tones interacting with a boundary tone.

However, not just any boundary tone will generate the complex array of alternations found in XT sandhi. The XT boundary tone is unique: it is a floating register feature (i.e. not a fully fledged tone but a subminimal tonal feature), and at the same time exhibits tonal polarity, surfacing with the value opposite to the register of the base to which it attaches. We call this a ‘polar register boundary tone’, abbreviated as $\textcircled{P}$ (for ‘register’). Polar boundary tones have been previously posited (Hyman 2007a), as have boundary tones consisting only of a register feature (Gjersøe et al. 2019). The XT boundary tone stands at the intersection of these two, logically predicted but to our knowledge not previously proposed.

In what follows, we demonstrate how a boundary tone-based analysis, coupled with the non-final-as-basic approach and a set of markedness constraints on tonal structure, derives *all* sandhi alternations in a natural way and in a manner which is strictly local. In other words, one can take an underlying tone and a boundary tone combination, and predict what the output will be for XT sandhi, with no need to appeal to global notions such as circles or chain-shifts. This obviates the concerns about the computability or non-naturalness of circles, since these are just epiphenomena.

Our paper is structured as follows. Section 2 provides an overview of tone sandhi in Sinitic and XT. Section 3 presents our analysis of the XT patterns, providing underlying representations, phonological derivations, and rules for mapping phonology to phonetics (i.e. surface pitch). Section 4 provides an addition argument for our proposal from the nominalizing suffix -a⁵² NOM, which shows a neutralization pattern exactly sensitive to the tonal features we propose. Section 5 provides a conclusion.

2. The Xiamen-Taiwanese tone sandhi

2.1. Background on Chinese tone sandhi

As a basic introduction to tone sandhi within Sinitic languages, we begin with Mandarin's well-known third tone (*T3*) sandhi.¹ Assuming Chao numbers for a pitch scale from 1 (the lowest) to 5 (the highest), the value of standard Mandarin third tone (*T3*) varies among three allotones [21], [214], and [35] (the sandhi form), distributed as in example (1). For the last allotone, the change to [35] creates neutralization with Mandarin's second tone (*T2*) which is underlyingly /35/.

- (1) a. [214]: found in phrase-final position; also in isolation, as this is always phrase-final, e.g. *ni*²¹⁴ 'you'
- b. [21]: found in non-final position, before any tone except for another third tone, e.g. *ni*²¹ *kan*⁵¹ 'you look'
- c. [35]: the result of dissimilation whereby a sequence 'T3 T3' becomes 'T2 T3' (sandhi), e.g. *ni*³⁵ *hao*²¹ *ma*⁰ 'you good Q?'

However, as an alternative to using Chao numbers this pattern can be understood within a bimoraic analysis of Mandarin with only L and H (and where the interrogative particle *ma*⁰ is toneless by virtue of being monomoraic). This is in (2).

- (2) a. [21] → LL: found in non-final position, before any other tone (except for another LL)
- b. [35] → LH: found as the result of dissimilation before another LL, whereby the sequence LL LL becomes LH LL
- c. [214] → LL($\textcircled{\text{H}}$): found in phrase-final position due to the contribution of an additional boundary tone ($\textcircled{\text{H}}$)

In (2), we can attribute the final H tone in phrase-final position as the contribution of a *boundary tone* ($\textcircled{\text{H}}$) – a prosodic element present at the end of every phonological phrase, but which is not associated with any lexical item, and thereby docks onto the last word of the phrase. The dissimilation rule in tone

¹As common throughout Sinitic, Mandarin tones are generally labeled with a tone number – i.e. Tone 1, Tone 2, Tone 3, and Tone 4 – an order which has no ontological status in the tonology of the language (any more than does the fact that D is the fourth letter of the alphabet).

sandhi neutralizes LL and LH while the post-lexical addition of the boundary tone ㄏ maintains the contrast. We conventionally circle the boundary tone to indicate that it enters the derivation unassociated to any particular host (i.e. it is floating).

Of the three allotonic realizations in (2), it is the /LL/ variant (i.e. [21]) which is underlying and specifically not the citation form which is underlying (nor the sandhi form for that matter). This conclusion has been reached in many theoretically-oriented works on Mandarin tone. Thus, Yip (1980: p. 319) asks “which of the two forms – citation or sandhi – is underlying”, and that “the traditional Chinese answer has always been to assume that the citation form is underlying, but there is no a priori reason why this should be the case”. Perhaps the earliest statement of this type in the English-language literature is Woo (1969: p. 201), who asserts that “it is not obvious that the isolation contours must be considered as lexical; recall the problems which arise in Mandarin phonology if the citation form of the third tone is taken to be lexical”.

Nonetheless, the trend for virtually all explanations of sandhi is to consider the isolation form as underlying (what Yip mentioned as “traditional” in the quote above). We explicitly reject this view, and assume that just like Mandarin third tone, Xiamen-Taiwanese tonal alternations are due to the contribution of a boundary tone. In a way, these common Mandarin facts pave the way for an understanding of related tone sandhi systems that is decidedly untraditional.²

2.2. Xiamen-Taiwanese tone contrasts

Xiamen-Taiwanese is our language of focus, part of the Sinitic sub-family known as Southern Min, spoken in parts of mainland China and Taiwan. There are numerous names for this dialect cluster, all of which essentially refer to the same language. These include Amoy (the exonym for the Chinese city *Xiàmén*), variations on Hokkien (e.g. Hagu, Hoklo, Holo) or Fujian (e.g. Fujianese, Fukienese), and variations on Taiwanese itself. In what follows, we shall collectively refer to the language as ‘Xiamen-Taiwanese’ (or ‘XT’). We will be explicit when data or analysis only pertains to a specific sub-dialect.

²Tsay (1994: p. 225-226) also draws a parallel between Mandarin third tone H addition and Xiamen-Taiwanese tone sandhi.

Otherwise, our review, description, and analysis extends to the language as a whole.³

There is consensus across the literature that XT has seven tones (the latter two being so-called checked tones only found in short syllables closed by an obstruent). In Table 1, we provide these seven tones in a near-minimal pair based on their pronunciations in isolation. For each, we provide its descriptive label, its conventional tone number (*T1-T8*), the traditional Chinese name for the tone category it descends from, an example written with XT conventional tone diacritics (note: not IPA), and the approximate pitch value for each tone from the studies compiled in Fon and Khoo (2024). We normalize over these sources (and the XT literature as a whole), and adopt the approximate phonetic values for this paper as provided in the rightmost column in Table 1.

³Due to space, we do not explicitly extend our analysis to the coastal Taiwanese variety described, *inter alia*, in Hsieh (1976) and Yip (1980). The main difference involves non-final realizations: the ‘33... ↔ 24#’ tone (*T5*) is realized as non-final [21] rather than non-final [33], and the ‘44... ↔ 52#’ tone (*T2*) is realized as non-final [33] rather than [44].

Table 1: Compilation of tonal realizations of seven tones in XT (Fon and Khoo 2024: p. 12)

	T #	Example	Chang (1989)	Yang (1991)	Tung (1996)	Ang (1997)	Chang (1999)	This paper
High-level	T1	(Yin Ping) <i>kun</i>	55	44	44	44	55	[44]
High-falling	T2	(Yin Shang) <i>kún</i>	53	53	53	41	51	[52]
Mid-rising	T5	(Yang Ping) <i>kún</i>	24	13	24	23	13	[24]
Mid-level	T7	(Yang Qu) <i>kūn</i>	33	33	33	33	33	[33]
Mid-falling	T3	(Yin Qu) <i>kùn</i>	21	31	11	21	11	[21]
High-short	T8	(Yang Ru) <i>kút</i>	53	33	4	4	5	[4]
Mid-short	T4	(Yin Ru) <i>kut</i>	21	32	32	32	3	[2]

Table 2: Final and non-final variants of seven tones in XT (Chen 2018: p. 5)

T #	Final	Non-final	Example	Final	Meaning	Non-final	Meaning
T1	44 ↔	33	<i>si</i>	‘poem’	[gim ³³ si ⁴⁴]	si ³³ jin ²⁴	‘poet’
T2	52 ↔	44	<i>sí</i>	‘death’	[thai ³³ si ⁵²]	si ⁴⁴ io ³³	‘dead end’
T5	24 ↔	33	<i>sí</i>	‘time’	[sui ³³ si ²⁴]	si ³³ kan ⁴⁴	‘time’
T7	33 ↔	21	<i>sī</i>	‘temple’	[hut ² si ³³]	si ²¹ bio ³³	‘temple’
T3	21 ↔	52	<i>sì</i>	‘four’	[te ²¹ si ²¹]	si ⁵² hong ⁴⁴	‘four sides’
T8	4 ↔	2	<i>sík</i>	‘ripe’	[bin ²¹ sík ⁴]	sik ² sai ³³	‘to know’
T4	2 ↔	4	<i>sik</i>	‘color’	[a ^{N21} sík ²]	sik ⁴ chhai ⁵²	‘color’

2.3. Previous analysis of Xiamen-Taiwanese tone sandhi

The form of these seven tones depends on their position within a tone sandhi group (for our purposes approximately the phonological phrase, but see Chen 1987). When these tones appear in final position, they surface as indicated in Table 1. However, if they appear in non-final position within the tone group they are traditionally described as taking a special sandhi form. Table 2 provides examples of each tone in both its final and non-final form, taken from Chen (2018) (we could also call these the citation vs. combination forms, respectively, using the terminology from Wright 1983).⁴

In Table 2, one can observe that the same morpheme surfaces as two different values taken from the inventory of tones, e.g. alternating between [44] and [52] or between [52] and [21], *etc.* Famously, these data reveal a chain shift when comparing the final form to the non-final form: 24 → 33 → 21 → 52 → 44 → 33. The circular nature of this chain shift is apparent in Figure 1 below, a common diagram across the XT theoretical literature. This tone circle (or tonal clock – Bodman 1955) shows that each sandhi transformation moves one step along an eventually circular path.

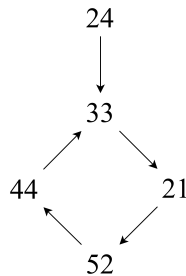


Figure 1: Commonly presented XT tone circle (Chen 2000: p. 42)

Many authors have highlighted the kind of ‘musical chairs’ effect of this circle (e.g. Chen 2000), which largely preserves lexical contrast across the entire system (e.g. Barrie 2006). Note, however, that we will reevaluate this tone circle later in our analysis, and see it ultimately as an incidental by-product.

⁴Note that in Table 2 we repeat the examples in the orthographic conventions provided by Chen (such as using capital ‘O’ for [ɔ]). Here and throughout, we unify the tonal transcriptions across sources but we do not seek to unify the segmental conventions.

The XT tone circle is traditionally understood as converting a final form (the starting point) to a derived non-final form (i.e. the change from final [24] to non-final [33] at the top of Figure 1). This assumes that the final form is its basic form (i.e. its underlying form in theoretical parlance), which is the received view adopted by the great majority of works examining the XT tone system (e.g. Chiu 1931, Chiang 1966, Wang 1967, Yip 1980, Chen 1987, Tsay and Myers 1996, Bao 1999, Moreton 1999, Chen 2000, Mortensen 2006, Barrie 2006, Thomas 2008, Hsiao 2015, Lin 2015, *inter alia*).

Aside from the present article, there have periodically been proposals that argue for the opposite in XT (and/or Southern Min more broadly): the non-final forms are basic and the final forms are derived, essentially reversing the direction of the tone circle in Figure 1 (e.g. Woo 1969, Ting 1982, Hashimoto 1982, Wright 1983, Tsay 1991, 1994). The final form is also the citation form of a word when spoken in isolation, and consequently these works reject the notion that the citation form of a word necessarily reveals its underlying form. To quote Hashimoto (1982), for example, “when a single syllable (morpheme) is uttered ‘in isolation’, it is not pronounced ‘in isolation’; it is in fact uttered as the final syllable of a phonemic phrase which happened to consist of a single morpheme” (p. 647).

Of these works, one particular proposal which is forceful, explicit, and detailed is Wright (1983). She presents a compelling argument for the non-final form as underlying based on a ‘retriplication’ process occurring with adjectives to express intensification. Example (3) acts as a baseline for comparison.

- (3) $thi\tilde{a}^{52}$ $thi\tilde{a}^{52}$ $thi\tilde{a}^{21}$ ‘very painful’

The analysis of such forms is that $thi\tilde{a}^n$ ‘painful’ ($T3$, and as rendered in its conventional transcription) is reduplicated twice: the first two copies are non-final and therefore surface as [52] while the last copy is in final position and surfaces in its citation form [21].

However, consider the retriplication process with *oai* ‘crooked’ in (4). Ordinarily, words of this tone category ($T1$) show an alternation between final [44] and non-final [33] forms (e.g. compare *phang* ‘fragrant’ in [chin phang⁴⁴] ‘very fragrant’ versus [phang³³ chui] ‘fragrant water’). In the case of retriplication, we would naively expect [33-33-44]. Nonetheless, there is a dissimilation rule in XT whereby the first (and only the first) instance of two surface [33] tones dissimilates to rising before another [33] in this

morphological context (what Wright 1983: p. 189 states “is essentially a variant of the by now familiar Mandarin Third Tone or Fuzhou Low Tone Dissimilation rule”).⁵

(4) oai²⁴ oai³³ oai⁴⁴ ‘very crooked’

What is happening in (4) is straightforward if these are underlyingly /33/. A sequence /33 33 33/ undergoes two rules: dissimilation of the leftmost element (and *only* the leftmost) within this morphological context (i.e. /33/ → [24]), and the addition of a right-edge boundary tone which changes the final copy (i.e. /33/ → [44]), reminiscent of the (H) boundary tone of Mandarin (e.g. (2) above).

Such data are problematic under the final-form-as-underlying approach, where the underlying sequence of (4) would be actually /44 44 44/. As Wright notes, in order to get both tones in non-final position to become [33] as well as to subsequently apply dissimilation and become [24], this would create level misordering, with phrasal rules (i.e. phrase-final i.e. /33/ → [44]) applying before compounding-level rules:

“If the combination form is only derived at the phrase level, then the phrase level rules will look down inside the word and perform melody distortion. This type of rule has been argued not to be possible in natural languages by a number of linguists.” (p. 189)

The problem of phrase-level rules feeding word-internal rules of dissimilation is only caused by the assumption that it is the citation final forms (i.e. those found in isolation) which are underlying. We therefore follow Wright’s conclusion (among the others cited above) that the non-final forms are actually the underlying ones, and that the actual sandhi rules only apply phrase-finally as the result of the addition of boundary tones. This in fact accords well with what various researchers have called the “demarcative function of Xiamen tone sandhi” (Chen 2000: p. 82). To quote Chen at greater length,

“The sandhi process is circumscribed by a domain that most closely approximates the p-phrase in prosodic hierarchy. . . Xia-

⁵Unfortunately, we do not have examples of retriPLICATION where the three copies are followed by another word within the same tone sandhi domain (none were found in the original sources, Cheng 1968, 1973). This would allow us to determine whether there is dissimilation between the second and third copies when the third is not phrase-final.

men p-phrasing functions like some sort of phonological punctuation: the appearance of the changed tone serves to signal the end of a major syntactic constituent" (p. 435 – our emphasis)

In this way, our analysis using boundary tones formalizes earlier statements whereby the juncture itself (i.e. the phrase boundary) is taken to be the conditioning environment (e.g. Tsay 1994: p. 225).

2.4. On the nature of Xiamen-Taiwanese tonal alternations

Looking back at the XT tone circle in Figure 1, numerous authors have attempted to find some kind of generalization which cuts across all the changes. Comparing the pairs, there is both contouring and decontouring, raising and lowering, and in the case of the short checked syllables, tonal exchange. The apparent phonological unnaturalness of these alternations holds whether one views the final or the non-final as underlyingly.

This situation has led many to conclude that XT tone sandhi is not derivable through classical derivational means in phonology (e.g. involving more-or-less uniform inputs and outputs). Such data were in fact the catalyst for the concept of “paradigmatic replacement” (Schuh 1978: p. 248), whereby one tone arbitrarily but systematically replaces another tone in a given environment (this is sometimes also called “arbitrary substitution” or “paradigmatic substitution” – Zhang 2007: p. 263). In fact, Moreton (1999) goes so far as to claim that XT sandhi is underivable and outside the scope of phonology proper:

“It looks as if the Taiwanese general tone-sandhi alternation is, synchronically speaking, completely arbitrary and idiosyncratic, not derivable from simple, logical, plausibly innate constraints, even if they are allowed to be non-conservative. It is not the kind of phenomenon OT was intended to explain, but a case of out-and-out paradigm replacement peculiar to Taiwanese.” (p. 36)

At the same time, questions of the psychological reality of XT sandhi further bolster incredulity over its phonological status (e.g. Hsieh 1970, 1976, Zhang et al. 2009). It should be emphasized, however, that some of the productivity studies suffer from potential methodological issues in creating tonal versions of words that already exhibit a tonotactic gap in the language. In contrast, one of the most ecologically valid studies is Chuang et al. (2011),

which found more promising results. This study places existing ‘cranberry’ morphemes in disyllabic compounds in isolation (using the head-final version of the expression ‘cran as in cranberry’, in Sinitic rendered as ‘cran-berry *de/e* cran’). This thereby places an *existing* morpheme into a sandhi context where it otherwise never appears. Chuang et al. (2011) found a higher rate of productivity than in previous studies, one “relatively higher than above chance (=20%), indicating that the general tone sandhi rules may not be so ‘unproductive’ if familiarity of the test words is enhanced” (p. 494).

In fact, the motivation behind many of these productivity studies is based on the view that the sandhi patterns are “characterized by circular opacity” (Zhang et al. 2011). However, if the sandhi patterns are in fact the result of phrase-final boundary tone addition, the circular pattern is simply epiphenomenal based on the changes that are possible within a limited set of tonal elements. We now turn to our analysis of XT tone sandhi (building on the insights of Woo 1969 and Wright 1983), and show the power of boundary tones in conjunction with underspecified tonal representations to derive all alternations.

3. Our Analysis

As stated, the goal of this paper is to derive all sandhi in terms of only the local addition of floating boundary tones. This stands in stark contrast to analyses such as Hsieh (1990), Barrie (2006), and Thomas (2008), that employ system-level constraints for the XT tonal system, whereby 5⁵ different potential sandhi patterns are evaluated as candidates.⁶ To this end, we must address (i) what are the underlying representations of the seven XT tones, (ii) how are the non-final forms derived from these representations, (iii) how are the final forms derived from these same representations via the addition of a boundary tone, (iv) what kind of boundary tone is this, and (v) how are the phonological representations mapped to phonetic values? We turn to these questions presently.

⁶See for example Thomas (2008: p. 428), who is explicit that “we will have to consider 3,125 candidates, the number of permutations with repetition of the sequence of possible tones”. Other authors implicitly evaluate all such candidates.

Table 3: Summary of seven XT tones decomposed into underlying features

T #	(Category)	Non-final	Final	Our featural analysis		
				1st μ	2nd μ	Register
<i>T1</i>	(Yin Ping)	33...	$\leftrightarrow$ 44#	H	H	<i>l</i>
<i>T2</i>	(Yin Shang)	44...	$\leftrightarrow$ 52#	H	$\emptyset$	<i>h</i>
<i>T5</i>	(Yang Ping)	33...	$\leftrightarrow$ 24#	H	$\emptyset$	<i>l</i>
<i>T3</i>	(Yin Qu)	52...	$\leftrightarrow$ 21#	H	L	<i>h</i>
<i>T7</i>	(Yang Qu)	21...	$\leftrightarrow$ 33#	H	L	<i>l</i>
<i>T4</i>	(Yin Ru)	4...	$\leftrightarrow$ 2#	H	-	<i>h</i>
<i>T8</i>	(Yang Ru)	2...	$\leftrightarrow$ 4#	H	-	<i>l</i>

3.1. Analysis part I: Underlying forms of Xiamen-Taiwanese tones

Recall the seven tones of XT (Table 1), which we summarize in Table 3 including its tone number and category for reference purposes, and its non-final and final value using our conventionalized pitch targets. The rightmost columns of this table summarize our featural analysis. We analyze all non-short syllables as composed of two moras (i.e. two tone-bearing units), but ‘checked’ syllables as composed of only one. All tones underlyingly have an initial H feature associated to the first mora, but they contrast as to which features dock to the second (if any).

The featural analysis in Table 3 acts as shorthand for a geometric representation of individual tones, exemplified in Figure 2. In this multi-tiered object, there are two types of features (as proposed, *inter alia*, by Yip 1980 and later Pulleyblank 1986, Snider 1999, Bao 1999). We follow Snider (1999) most closely, placing one set of features on a ‘tonal tier’ which oppose [H] and [L] values, and place register features on a ‘register tier’ which divides the pitch space into two broad regions, a higher [*h*] and lower [*l*] region. These features are on separate tiers, but are connected via a ‘tonal root node’, which then associates to a tone-bearing unit (for us, the mora μ). Thus, the pitch space is partitioned into registers *h* and *l*, and within a register an H is realized towards the higher limit of that register and an L towards the lower limit of that register. This may result in tones of different registers phonetically ‘overlapping’ in the pitch space (despite being phonologically distinct), part of our analysis to follow.

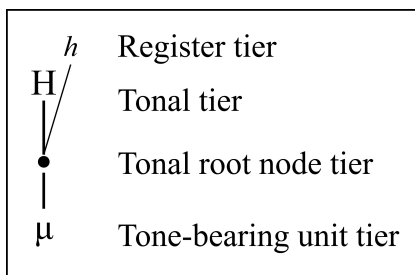


Figure 2: Geometric representation of tone features

Having established the parameters of our feature geometry, we provide the featural representations of all seven tones in Figure 3. This is divided into a 2x4 table where the columns are contrastive configurations of [H] and [L], while rows are contrastive register features [*h*] and [*l*]. In all tones, the first mora always bears a [H] tonal feature and there is no variation in this regard. Furthermore, there is only one register feature per unit, and this always associates to the first tonal root node and never the second.

It is on the second mora where we to see contrast. The leftmost column of Figure 3 shows the second mora unspecified for any tonal feature – this strictly gets its value via various tonological processes. Here, the tonal root node only associates to a single mora. We can contrast this to the next column, where the tonal root node spreads to both moras. This representational contrast will become important as it accounts for the distinct phonological behavior of tones *T1* vs. *T5* despite the fact that they both surface as [33] in non-final position. There is a gap in the high register here, which at this point we interpret as incidental. Moving rightward, in the next column the second mora bears a [L] tonal feature (i.e. it is an underlying contour, unlike all other tones). Finally, the last column constitute the short checked syllables with only one mora. These bear the only possible initial tonal feature [H] and either [*h*] or [*l*].

In total, XT underlying representations have the following restrictions:

- (5)
 - a. All representations begin with a tonal feature [H]
 - b. Tonal features only associate to a single root node
 - c. There is only one register feature per tone
 - d. Register features only associate to the first root node
 - e. A root node only associates to one feature per tier, i.e. one

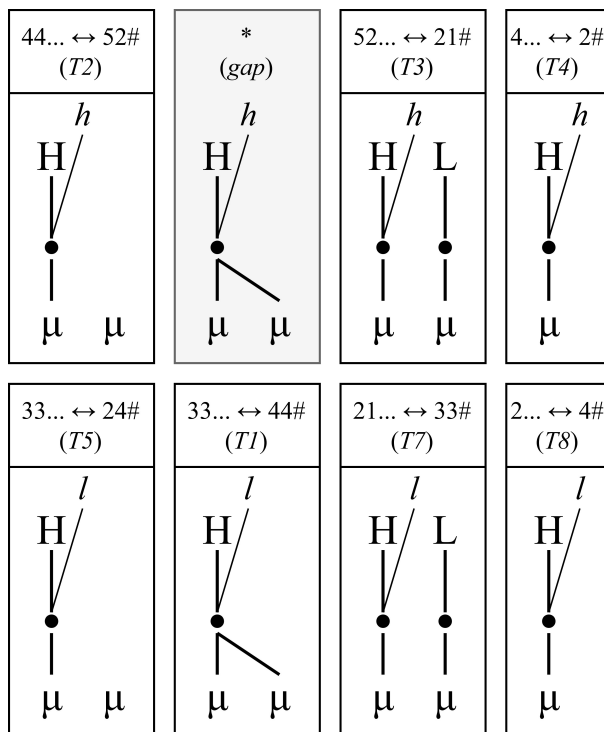


Figure 3: Underlying representation in features of the seven XT tones

tonal feature and one register feature), though it may associate to multiple moras

While having restrictions on underlying representations runs counter to common assumptions in Optimality Theory – i.e. the Richness of the Base (Prince and Smolensky 1993) – in recent years, this has been questioned in the phonological literature. This actually returns to an older position advocating for Morpheme Structure Constraints (or MSCs – Halle 1959), which impose phonological restrictions on the lexicon which are separate from constraints enforced in the grammar (for recent discussion, see Gouskova 2023, Rasin 2023). Our proposal is in line with this position.

3.2. Analysis part II: Deriving non-final forms

Having established the underlying representations for XT tones, we must now derive their output phonological representations in the two relevant contexts: the non-final and final position of a tone group. Neither the non-final nor the final set of forms is representationally identical to the underlying forms across the board, though the non-final forms are clearly much closer.

Deriving non-final forms is relatively straightforward and only involves specifying underspecified material. We may assume a well-formedness condition in non-final forms that all features and all moras must be associated to a root node, and all root nodes must be associated to a tonal feature, a register feature, and a mora. No underspecified or underassociated structure is allowed in the output. Formally, this can be enforced via a modified version of the constraint SPECIFYT (Yip 2002: p. 83) stating that “a TBU must be associated to a tone” which we interpret as the requirement that a tonal root node be associated to a tonal feature on the two tiers.

For three of the tones from Figure 3, this is incidentally already satisfied in the underlying representation. These tones are shown in Figure 4, showing no input-output changes, and surface as [33], [4], and [2], respectively. Notice that although [33] and [2] are phonetically distinct, they both involve a [H,l] root node phonologically. We return to the mapping of our phonological outputs to phonetic values in Section 3.4.

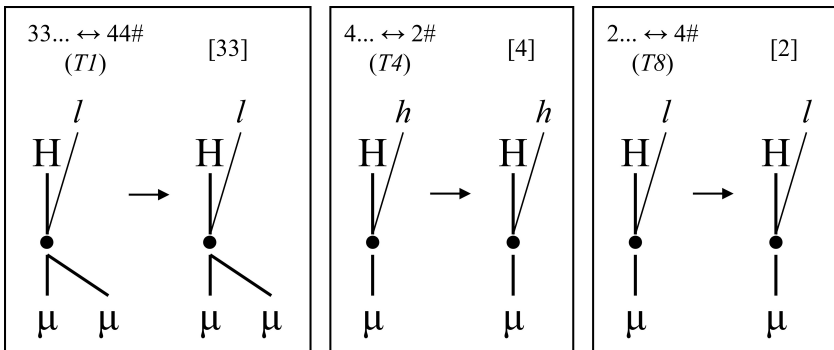


Figure 4: Deriving non-final forms – No change from input to output

In contrast, the remaining tones show underspecification. If a tonal root node is unspecified for a register feature, then neighboring register spreads

onto it, shown in Figure 5. The result is a falling contour at the high register [52] and low register [21], respectively. Moreover, if a mora is unspecified then the tonal root node spreads onto it, shown in Figure 6. This results in level [44] and [33], in two different registers respectively. It is with these tones that the constraint SPECIFYT becomes crucial.

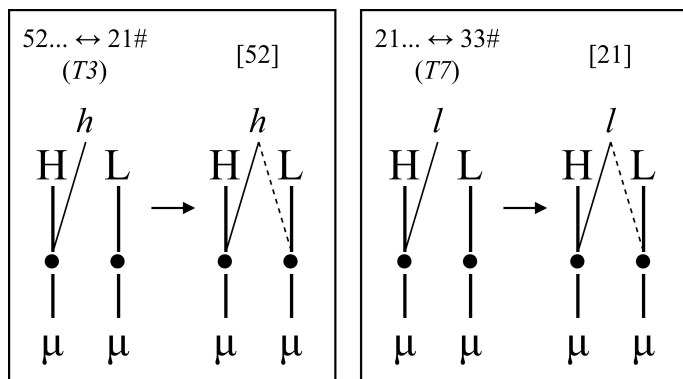


Figure 5: Deriving non-final forms – Register feature spread in output

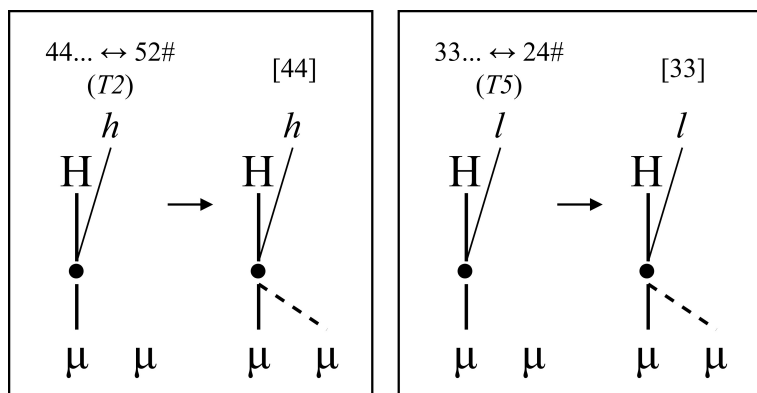


Figure 6: Deriving non-final forms – Root node spread in output

3.3. Analysis part III: Deriving final forms via polar boundary tones

In this section, we derive the final forms from the proposed underlying representations. It is here that we deviate most noticeably from previous authors but with what we believe is good cause: the tonal alternations previously characterized as “underivable”, “arbitrary”, and “unnatural” can be captured via a simple and independently posited mechanism, a floating *boundary tone* at the end of a phrase.⁷ This proposal turns sandhi on its head: tone sandhi actually takes place phrase-finally and *not* phrase-medially, taking seriously Chen’s quote above of XT tone sandhi as phrase-final punctuation.

Unlike the boundary tone sketch for Mandarin in (2), the boundary tone in XT has two unique properties: the boundary tone consists only of a register feature, and this register feature polarizes based on the register feature at the edge of the phrase. Polar boundary tones have been previously proposed in a distantly related Sino-Tibetan language Thlantlang Lai, for which Hyman (2007a) states:

“any final level tone must acquire a following [-αT] ‘polar boundary tone’: H% after final /L/, L% after final /H/, where % = pause”

The H% and L% boundary tones proposed by Hyman correspond roughly to our tonal features [H] and [L]. Several other studies have similarly shown boundary tone sensitivity to its phonological environment, which can change the value of the boundary tone. In Seoul Korean (Jun 1998), accentual phrases are marked automatically by a [LHLH] or [HHLH] intonational sequence, where the initial tone can be understood as an initial boundary tone which surfaces as H or L “determined by the laryngeal feature of the phrase-initial segment” (p. 193), such as aspiration or tenseness.

At the same time, boundary tones have been proposed which consist only of register feature, e.g. a low register boundary tone /% in the Niger-Congo language Limbum at the Cameroon/Nigeria border (Gjersøe et al. 2019). XT represents a unique intersection of these types: a polar register boundary tone.

⁷Wright (1983) also posits floating tones in her analysis. The key difference is that for Wright, floating (H)’s and (L)’s are part of the underlying representation of the seven tones themselves. In contrast, we take them to be purely a product of the prosodic environment (i.e. boundary tones at phrase edges).

We abbreviate this polar register tone as $\textcircled{r}$, but shall transcribe it as $\textcircled{h}$ or $\textcircled{l}$, respectively, when we place them in phonological derivations.⁸

Let us show how a polar register boundary tone derives the attested final forms. First, consider the tones whose second mora is underlyingly unspecified for tonal features, in Figure 7. Here, the floating polar register boundary tone takes the opposite value of the base. In this situation, the boundary tone's root node and the second mora meet each other's markedness needs (recall SPECIFYT), and associate to one another (the dashed lines in the output). The root node also needs to associate to a tonal feature, and thus [H] spreads. The addition of the boundary tone creates a derived register contour which is otherwise disallowed. Thus, otherwise level [44] and [33] become falling [52] and rising [24], respectively.⁹

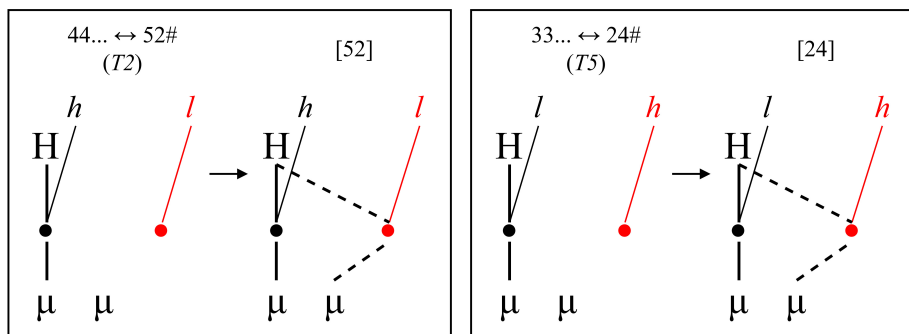


Figure 7: Deriving final forms – Association of floating root node

Next, consider the two tones with underlying [HL] where the [L] tonal feature is associated to the second mora, in Figure 8. In this situation, the

⁸For the purposes of this paper, we remain neutral as to the precise mechanics of tonal polarity. We may tentatively assume a high-ranked constraint against phrase-final OCP violations on the register tier, which results in obligatory dissimilation (see, *inter alia*, Trommer 2016 for the role of the OCP in tone polarity). Under this scenario, the underlying value of the floating boundary tone could be either $\textcircled{h}$ or $\textcircled{l}$ without consequence. We render it as $\textcircled{r}$ both as a useful mnemonic and to remain neutral as to this aspect of the representation.

⁹A reviewer asks about the structure of the floating boundary tone: is it a feature associated to a tonal root node or a more economical representation consisting only of a floating feature without such a root node. We discern no empirical difference between these positions, and maintain the former representation under the assumption that it is highly marked for a feature at any stage in the derivation to be unassociated to an organizing root node.

second mora comes in pre-associated, which is maintained in the output. In order for the register feature of the boundary tone to associate, it delinks from its own root node (indicated by crossing through its association line and making it opaque) and re-associates to the base final root node. We can assume a constraint *FLOAT which requires a tonal root node associate to a TBU and when this is not possible the root node deletes. A parallel *FLOAT which specifically references features instead of root nodes would require it to re-associate to a remaining node.¹⁰

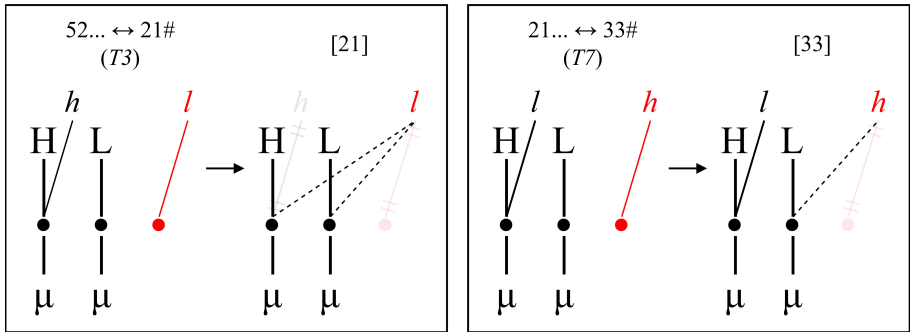


Figure 8: Association of boundary register feature to final root node of base

Consider the first tone in Figure 8 – the ‘52... ↔ 21#’ (T3) – where the docking of the boundary feature $\textcircled{l}$ creates a [L,l] unit adjacent to a [H,h] unit. This phonological configuration is not found within the XT system as a whole, and we therefore conclude that it is in fact categorically banned (both underlying and derived). A simply *[H,h][L,l] constraint prohibiting this sequence would suffice. To repair this structure, the boundary tone $\textcircled{l}$ docks to both the first and second root nodes (and the otherwise [52] becomes [21] with a total register change).

Next consider a different complication which arises with the second tone of Figure 8 – i.e. T7, ‘21... ↔ 33’. Here, we see that both the tonal features and the register features are opposite (i.e. [H,l][L,h]), but despite this opposition the whole sequence surfaces as [33] (i.e. level mid). Here, we follow previous work across Sinitic which exploits the possibility that level mid pitch (i.e. [33]) may

¹⁰A reviewer asks for T3, why the output is [H,L,l] (surfacing as [21]) with a low register feature, rather than [H,L,h] (surfacing as [52]) with a high register. We speculate that there is a constraint that forces the boundary tone to associate when it is present.

correspond to distinct phonological representations. For example, the overview of tonal systems across Sinitic in Bao (1999) reveals a number of cases in which mids are representationally ambiguous, where the surface appearance of a mid within a single inventory is in fact the result of two phonologically underlying distinct combinations, either [H,*l*] or [L,*h*]. Proposals involving two phonologically distinct mid tones are common outside of Sinitic, as well. One example is Trommer (2021) who analyzes tonal changes in roots in Gaahmg as involving a tonal inventory of H, M1, M2, and L, where the two mid tones are phonetically identical but possess different tonal features to capture distinct phonological behavior. For further theoretical discussion of the representation of mid tones, see van Hugte (this volume).

We contend that a level mid in a complex toneme (e.g. involving two moras) may also result from their combination, i.e. the [H,*l*][L,*h*] sequence in Figure 8. The ambiguity of level mids may help to explain the XT tone circle (Figure 1 above), which in actuality is not circle-shaped but rather ‘lollipop’-shaped, in the sense that one particular tonal node is the input/output of two distinct sandhi patterns. The hub of this lollipop is [33] (which alternates with both [24] and with [55]). We contend that it is no accident that for lollipop-shaped sandhi patterns, the hub is at the representationally ambiguous part of the inventory. We return to this discussion shortly in Section 3.4.

Finally, the last three tones we need to derive are in Figure 9. With these tones, there is only a single root node and all moras exhaustively associate to this root node underlyingly. As in Figure 8, the polar register boundary tone has no available mora to dock to and therefore it delinks from its own root node and re-associates to the base’s root node. This results in delinking of the underlying register feature, as no tonal root node can ever bear two register features. This appears to be a categorical prohibition, for which we can posit a constraint ONEREG/TRN (where TRN means ‘tonal root node’), which states that there can only be one register feature per root node (this modifies the constraint ONET/ μ from Yip 2002). Linking one register feature and delinking the other results in the exchanging quality seen in particular with the latter two tones (those on the short checked syllables).

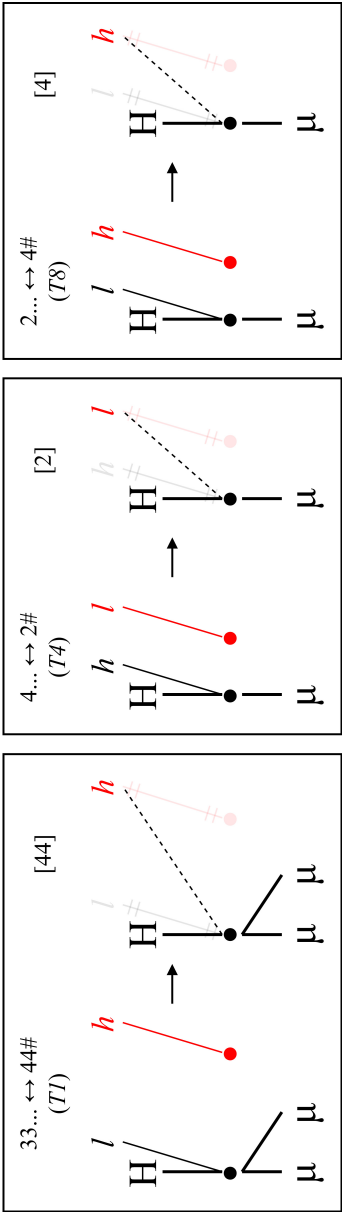


Figure 9: Deriving final forms – Boundary register features overwrite that of a single root node

In summary, based on our underlying representations, the final forms are derivable through a single addition in the input (in the spirit of what Tsay 1994 calls a ‘single process’), namely a polar register boundary tone, which surfaces as $\textcircled{h}$ (high register) or $\textcircled{l}$ (low register) depending on the register of the base. The attachment of this boundary tone results in spreading rules to repair marked structures, e.g. moras, root nodes, and features which are associated to too many or too few counterparts on different tiers. Our analysis successfully derives all surface patterns, using polarity to capture the exchanging quality of many of the tones when comparing their non-final to final forms (a pattern sometimes called ‘flip-flop’ in the sandhi literature – Wang 1967, Yue-Hashimoto 1986). The end result may still be considered a tone circle on the surface, but this is a purely epiphenomenal effect.

3.4. Analysis part IV: Phonology-phonetics mapping

The last component of our analysis is providing the rules which map the final phonological representations to their surface pitch values. This will be based on the specific tonal root node (again, TRN) associated to a mora. There are four types of TRNs, which differ in terms of their featural content: $[H, h]$, $[H, l]$, $[L, h]$, and $[L, l]$. The surface pitch they map to depends on their precise environment, but they all fall into a specific pitch range. This is provided in Figure 10, which uses the traditional Sinitic 5-pitch range (where 5 is highest and 1 is lowest). Moras associated to a $[H, h]$ are mapped to a [3-5] pitch range which we designate as a superscript ‘ $4\uparrow$ ’ with a vertical arrow in this figure (the mapping from phonology to phonetics is denoted by the dashed arrow). Moras associated to either of the TRNs $[H, l]$ or $[L, h]$ (i.e. the ‘mid’ tones) are mapped to a [1-3] pitch range, designated ‘ $2\uparrow$ ’. Lastly, moras associated to $[L, l]$ are consistently mapped to [1] (the lowest pitch).

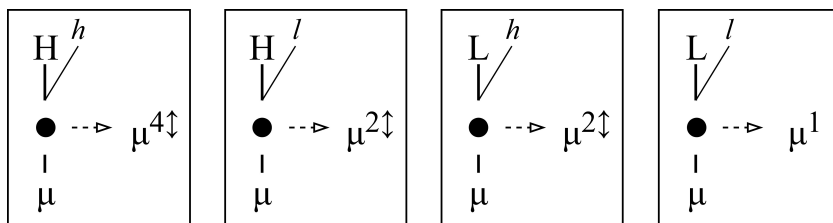


Figure 10: Mapping from phonology to phonetics of TRN-associated moras

How these rules map the phonological to phonetic forms of the seven XT tones is shown in Figure 11. The phonology-phonetics mappings of the tones when they appear in non-final position are in the left column while mappings of tones in phrase-final position are shown at the right. In this rightmost column, the red portion is the polar register boundary tone ($\hat{r}$), surfacing with the opposite register of its base. At the phonology-phonetics interface, each mora associated to a TRN is uniformly mapped to a pitch range according to the rules of Figure 10. Once this mapping is completed, the phonetic grammar makes adjustments to this pitch range depending on its context. In Figure 11, these adjustments are denoted with a curved arrow (distinct from the dashed arrow which indicates the phonology-phonetics interface).

Specifically, moras with ‘4↑’ are realized as a [4] pitch by default but as a special higher [5] pitch before a [2]. This is seen in both the non-final form of *T3* and the final form of *T2*. Pitch expansion of falling pitch sequences is extremely common cross-linguistically, what Hyman (2007*b*) refers to as ‘ f_0 polarization’. For example, in Igbo high tone is realized at a noticeably higher pitch before a low tone than with another high or following a low tone (Hayes 2009: p.292). That is, /H-H/ in Igbo using Chao numbers would be something like [3-3], /L-H/ would be [2-4] and /H-L/ would be [5-2] or even [5-1]. Similar findings are found across the world (e.g. in Thai, Yoruba, Engenni, Mankon, Moba, Krachi, and Kirimi, to name a handful of languages – see Hyman 2007*b* for details and references).

Returning to Figure 11, moras with ‘2↑’ are realized as [2] pitch by default, but a special phonetic adjustment is made: if two ‘2↑’ moras are adjacent, they are realized as a special higher [3] pitch. Finally, moras with ‘1’ are consistently realized as [1], the lowest pitch. No other phonetic adjustments are required.

As a final note on the phonology-phonetics mapping, the reader should recall that these surface pitches are themselves abstractions drawn from the literature, which we intend as a compromise between various descriptions. We reported back in Table 1 variation in the surface pitch of the seven XT tones taken from the survey in Fon and Khoo (2024), e.g. the rising tone of *T5* realized as either [13], [23], or [24] depending on the study. A fully comprehensive study of XT tone sandhi would need to account for this variation in a more thorough way than presented here. At the same time, we contend that our phonological representations and even our phonology-phonetics mapping procedure is uniform and could be applied to any of these studies. This

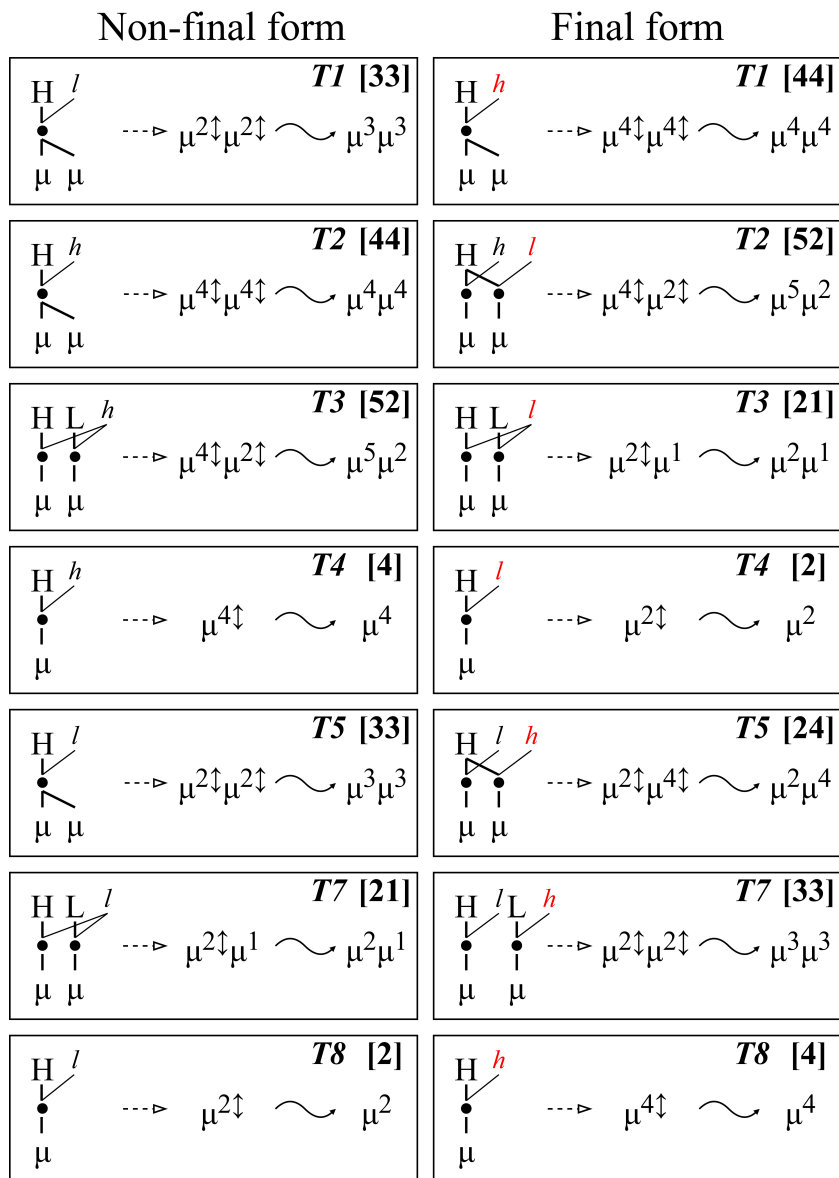


Figure 11: Complete set of mappings with adjustments from phonetic grammar
 (Note: the polar register boundary tone is in red)

provides the flexibility required for implementation in the phonetic grammar involving individual and real-time variation.

4. Additional support: Register maintenance with *-a* suffixation

An additional piece of evidence in support of our analysis involves the suffixation of a nominalizer/diminutive marker *-a*⁵², which we shall hereafter gloss simply as NOM (on the semantics and further exemplification, see, *inter alia*, Cheng 1968, Yip 1980, Wright 1983, Hsieh 1990, and Fon and Khoo 2024). When this affix is added, the tonal contrast of the base partially neutralizes, shown in Table 4. Tones *T1*, *T5*, *T7*, and *T8* neutralize and become a level mid pitch [33] or [2] if checked (Table 4a-e), while tones *T2*, *T3*, and *T4* become a level high-tone [44] (Table 4f-i).

Under our analysis, this distribution is not arbitrary: those tones which have an underlying [*l*] register surface with as lower [33]/[2] tone while those with underlying [*h*] surface as higher [44]. Thus, while [H] and [L] features neutralize in this context, register [*h*] vs. [*l*] is maintained. Here, we build on an earlier statement in Yip (1980) who writes that “before the suffix [*-a*⁵²], the Tones of a morpheme are deleted but its Register remains behind” (p. 322).

Table 4: Partial neutralization of seven tones in context of -a⁵² NOM (Fon and Khoo 2024: p. 15)

T#	NOM	Example	UR	Our analysis	Non-final	Final	NOM -a ⁵²	Meaning
a. <i>T1</i>	→	<i>kim</i>	/kim/	H H	[kim ³³ ...]	[kim ⁴⁴ #]	[kim ³³ -a ⁵²]	‘gold’
b. <i>T5</i>	→	<i>iunn</i>	/ĩũ/	H Ø	[ĩũ ³³ ...]	[ĩũ ²⁴ #]	[ĩũ ³³ -a ⁵²]	‘sheep’
c. <i>T7</i>	→	<i>phōo</i>	/p ^h ɔ/	H L	[p ^h ɔ ²¹ ...]	[p ^h ɔ ³³ #]	[p ^h ɔ ³³ -a ⁵²]	‘booklet’
d. <i>T8</i>	→	<i>hióh</i>	/hiəʔ/	H -	[hiəʔ ² ...]	[hiəʔ ⁴ #]	[hiəʔ ³³ -a ⁵²]	‘leaf’
e. <i>T8</i>	→	<i>tshát</i>	/ts ^h at/	H -	[ts ^h at ² ...]	[ts ^h at ⁴ #]	[ts ^h at ² -a ⁵²]	‘thief’
f. <i>T2</i>	→	<i>láng</i>	/laŋ/	H Ø	[laŋ ⁴⁴ ...]	[laŋ ⁵² #]	[laŋ ⁴⁴ -a ⁵²]	‘cage’
g. <i>T3</i>	→	<i>thòo</i>	/t ^h ɔ/	H L	[t ^h ɔ ⁵² ...]	[t ^h ɔ ²¹ #]	[t ^h ɔ ⁴⁴ -a ⁵²]	‘rabbit’
h. <i>T4</i>	→	<i>ah</i>	/aʔ/	H -	[aʔ ⁴ ...]	[aʔ ² #]	[a ⁴⁴ -a ⁵²]	‘duck’
i. <i>T4</i>	→	<i>tik</i>	/tik/	H -	[tik ⁴ ...]	[tik ² #]	[tik ⁴ -a ⁵²]	‘bamboo’

Our analysis of this NOM morpheme is that it underlyingly bears a complex contour [HL] in a high register [*h*], but crucially this also sponsors a defective tonal root node before it.¹¹ This root node is associated to the [H] feature but has no register feature of its own, and is not pre-associated to any mora. Our proposed underlying representation is in Figure 12.

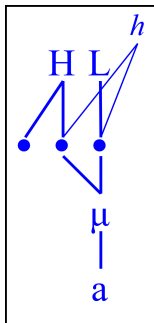


Figure 12: Underlying representation of nominalizer -a⁵² NOM

In context, the final mora of the base associates to the defective tonal root node, which causes the base mora to delink from the root node it is underlyingly associated to, which results in the loss of a tonal feature [H] or [L]. However, this erstwhile defective root node requires register, and the register of the base survives on this newly added root node. Representative examples of the [HL] tone (*T7* and *T3*) and checked [H] tones (*T8* and *T4*) are in Figure 13, showing both registers.¹²

¹¹It is unclear whether this suffix should be analyzed with one or two moras, and we follow the general convention in the literature involving a single vowel and single mora. Thus, both tone features [H] and [L] exceptionally dock to a single mora (otherwise disallowed).

¹²The reader may notice a discrepancy: in Figure 13 with this suffix -a⁵², the defective root node associates to the base mora, and its underlying root node then delinks. In contrast, with the polar register boundary tone in comparable environments, it is the boundary tone's root node which deletes and the base mora remains associated to its underlying node (cf. Figures 8 and 9). As we saw in the Mandarin case (1), this difference in association in XT is one of several places where boundary tones obey stricter faithfulness constraints when they combine with a base than tonal structures which do not involve boundary tones. We leave a more explanatory account of these XT facts for future work.

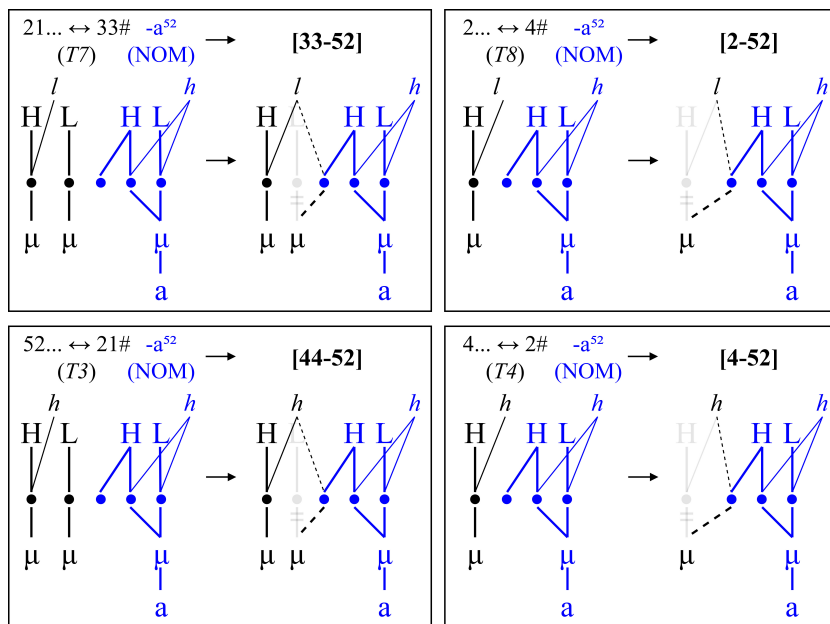


Figure 13: Maintenance of the base's register with suffix $-a^{52}$ NOM

5. Conclusion

We have reexamined tone sandhi in Xiamen-Taiwanese, including its famous tone circle whereby tones appear to rotate in value. We demonstrated the seven-way tone contrast in Xiamen-Taiwanese, where each tone has two realizations depending on its position in a tone sandhi group, e.g. tone *T1* (i.e. Yin Ping) is realized as a level mid pitch [33] in non-final position but as a level high pitch [44] in final position. Building on analyses of phrase-final Mandarin tonal alternations involving boundary tones, we have exploited the possibility of boundary tones in Xiamen-Taiwanese to account for comparable alternations. Our analysis goes against the traditional Chinese analysis and sees the non-final forms as the base forms (roughly, the underlying forms) while the final forms are derived (building on the earlier work of Woo 1969 and Wright 1983). Thus, in the oft-cited 12-syllable example from (Chen 1987: p. 113), our analysis only involves altering the base tone of the final syllable, in bold in (6).

- (6) [44 24 24 21 52 44 21 21 52 44 52 **2**]
 yi kiong kiong kio gua ke k'uah puah tiam tsing ku **ts'eq**
 (he by force cause I more read half hour long book)
 'He insisted that I read for another half an hour'

In contrast, in the traditional analysis tone sandhi applies to all non-final syllables, amounting to eleven syllables in a row. These are in bold in (7).

- (7) [**44 24 24 21 52 44 21 21 52 44 52 2**]
 yi **kiong kiong kio gua ke k'uah puah tiam tsing ku ts'eq**

Considerations from language acquisition and speech planning (along the lines of Wagner 2012) would favor such a re-analysis, where it is only the last syllable that is changed and requires a special rule.

Specifically, we have posited that the seven tones of Xiamen-Taiwanese should be decomposed into features. For all tones, the first mora bears a [H] tonal feature, while the second mora bears either this same [H] feature (i.e. doubly-linked), a [L] feature, or is toneless underlyingly (checked syllables have only one mora). Each of these configurations then bears either a high register feature [*h*] or low register [*l*], leading to eight potential underlying forms (there is an incidental gap of [H] pre-associated to two moras with high register [*h*], thus resulting in seven tones).

From these underlying representations, non-final forms surface faithfully (*modulo* certain markedness-driven changes involving spreading to underspecified moras and root nodes). In contrast, final forms bear a polar register boundary tone which appears at the end of every tone sandhi group, which incidentally includes words spoken in isolation. If the group ends underlyingly in high register the boundary tone is [*l*] but if it ends in low register the boundary tone is [*h*]. We called this a 'polar register boundary tone' or (*r*), which is logically predicted to occur but prior to this study had not yet been identified in the world's languages.

The polar nature of this boundary tone results in the largely structure-preserving patterns exhibited by the tone sandhi system, where there is rampant change but no neutralization of contrast. Thus, Xiamen-Taiwanese tone sandhi is far from arbitrary, underivable, or unparalleled, but rather constitutes a well-behaved system using familiar notions of underlying contrast, tonal features, boundary tones, polarity, and markedness repairs.

The next steps of this analysis would be to extend it to other Southern Min

varieties, such as the Coastal Taiwanese dialect or Teochew/*Cháozhōu*, which show comparable sandhi alternations. More broadly, these results call out for a broader investigation into boundary tones, polarity, and their interface. There has been progress of late on polarity effects with separate register and tonal nodes (Meyase 2021), and Tebay (this volume) adopts tonal circumfixes for Igbo which affect the last element in NP in a way that suggests boundary tones (with potentially parallel effects in Nilotic languages Jumjum and Kalenjin). All of this clearly points to the need for investigating segmentally invisible elements in phonology, detectable through their effects on the surrounding environment.

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